Computer Graphics

CH1: Introduction to Computer Graphics

Outline

- **What is Computer Graphics?**
- **Components of Computer Graphics**
- **Application Areas of Computer Graphics**
- **Graphical Display Devices**
- **Graphical Input Devices**
- **Software Standards – Open GL, GKS, PHIGS**

What is Computer Graphics?

- **Computer graphics** generally means *creation*, *storage* and *manipulation* of *models* and *images.*
- Computer graphics is the process of creating an object into a computer.
  - Using a computer as a rendering tool for the generation (from models) and manipulation of images is called computer graphics.
- It is about how to program a computer to generate photo-realistic images.
  - Technically, it's about the production, *manipulation* and *display* of images using computers; and
  - Practically, it's about movies, games, art, training, advertising, communication, design, etc.
- Generally ***Computer graphics*** deals with graphics created using computers and the representation of image data by a computer specifically with help from specialized graphic hardware and software.

Components of Computer Graphics

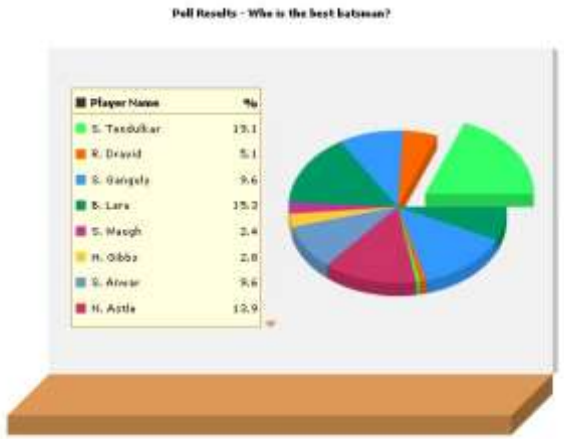
- Computer graphics deals with all aspects of creating images with a computer – *Hardware*, *Software* and *Applications*.
- **Modeling**:
  - Modeling refers to the process of defining objects in terms of primitives, coordinates and characteristics and creating and representing the geometry of objects in the 3D world.
- **Storing**:
  - Storing scenes and images in memory and on disk.
- **Manipulating**:
  - The process of changing the *shape*, *position* and *characteristics* of objects in a scene.
- **Rendering**:
  - The process of applying physically based procedures to generate (photorealistic) images from scenes (using lighting and shading).
- **Viewing**:
  - The process of displaying images from various viewpoints on various devices.
- **Animation**:
  - (movement) describing how objects change in time.

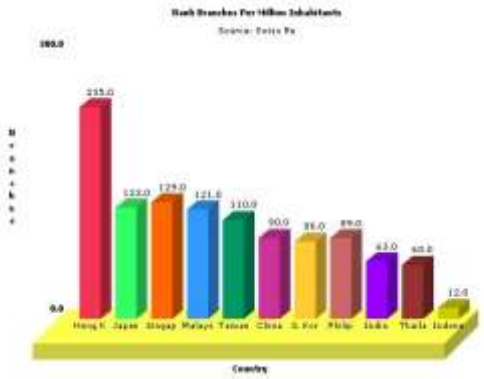
Application Areas of Computer Graphics

- Presentation Graphics
- Computer Aided Design(CAD)
- Visualization
- Computer Art
- Education and training
- Image processing
- Entertainment
  - Movies Industry
  - Gaming Industry
- Medical field
- Graphical User Interface(GUI)

Application Areas of Computer Graphics Presentation Graphics

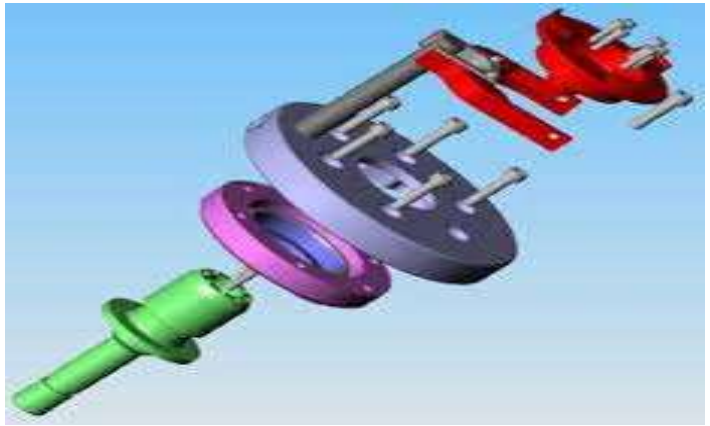
- Used to summarize the *financial*, *mathematical*, *scientific* and *economic* data.
  - Typical examples are bar charts, line graphs, pie charts etc.

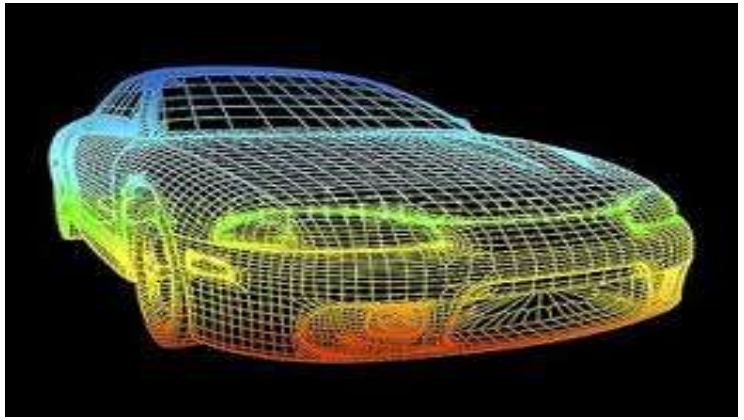


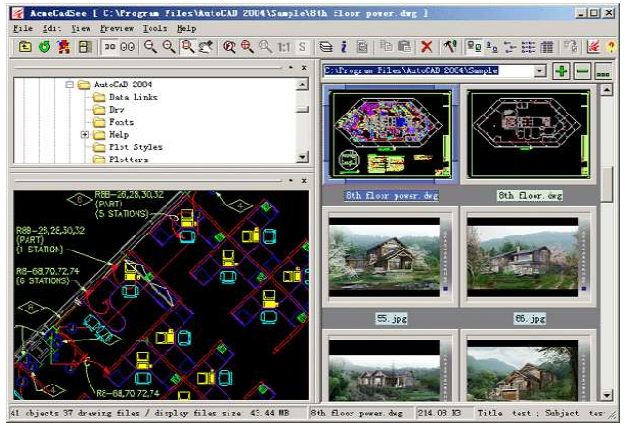


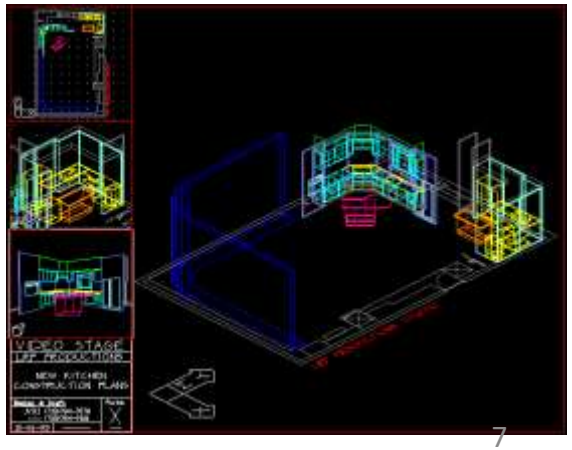
Application Areas of Computer Graphics Computer Aided Design(CAD)

- Major use of computer graphics is in design process, particularly for engineering and architectural systems.
  - This include design of buildings, automobiles, aircraft etc.



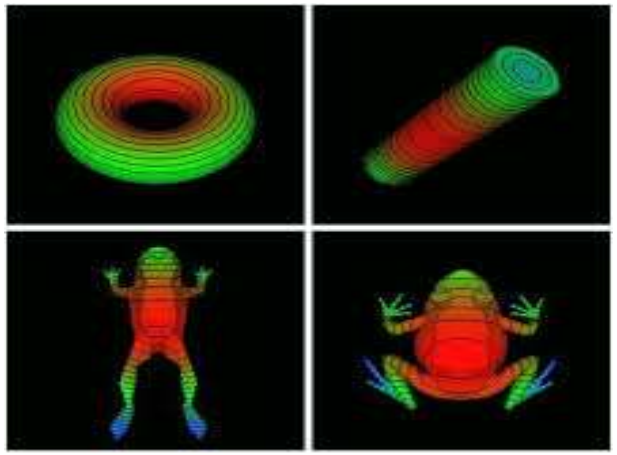


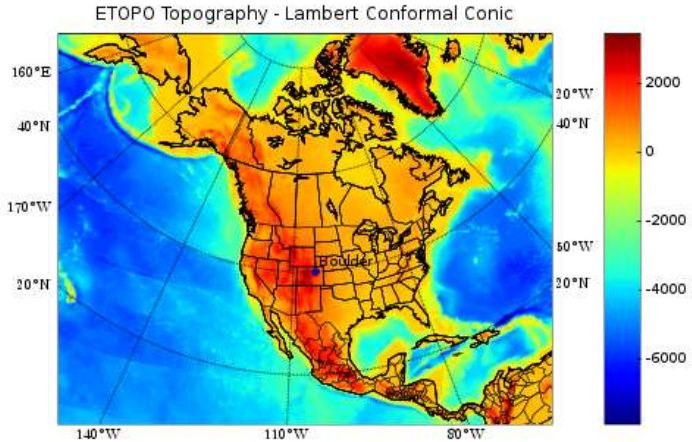


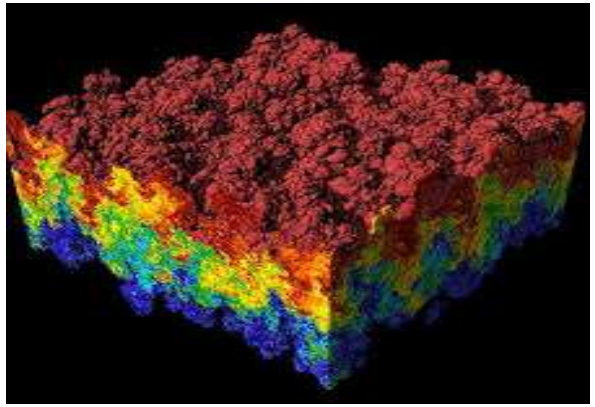


Application Areas of Computer Graphics Visualization

- Various techniques can be used to represent the large amount of data obtained from scientific, medical or business analysis.
  - These includes color coding, contour plots, graphs, charts etc.







Application Areas of Computer Graphics Computer Art

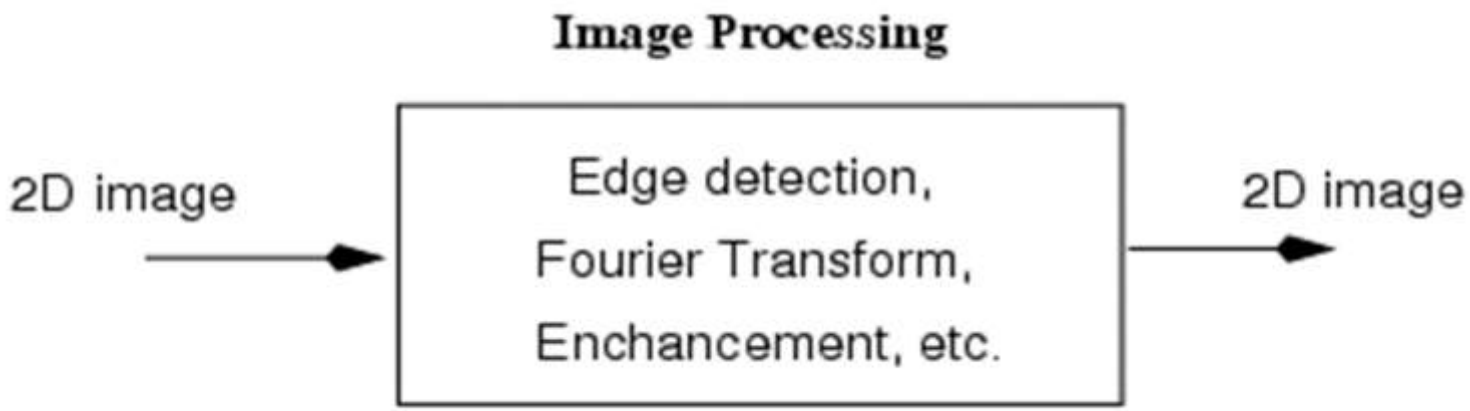
- Artist uses special purpose hardware and programs that provides facilities for designing object shapes and specifying object motion.
  - Examples pixel paint, super paint etc.

Application Areas of Computer Graphics Education and training

- Computer generated models of physical, financial and economic system are often used as educational aids.
  - Various kinds of simulators program can be used to provide the trainings.
  - E.g. automobile driving simulator.

Application Areas of Computer Graphics Image processing

- Computer graphics is used to create pictures.
- Image processing applies techniques to modify or interpret the existing pictures. It is used to:
  - Improve picture quality
  - Machine perception of visual information

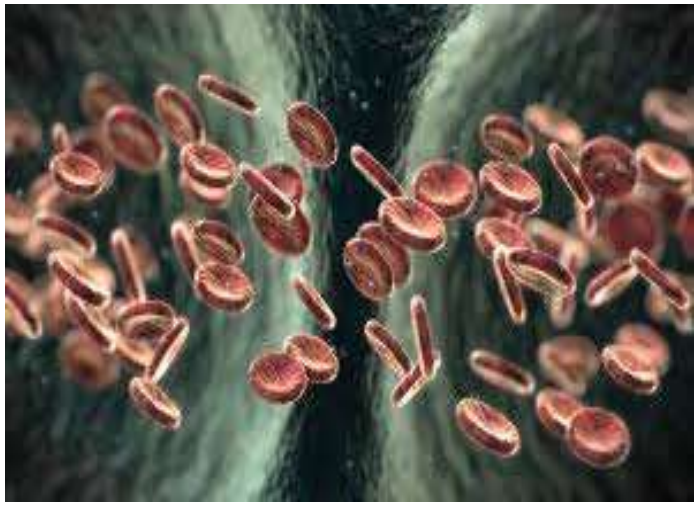


Application Areas of Computer Graphics Entertainment

- Computer graphics methods are now commonly used in making motion pictures, music videos , games and televisions shows.
  - Sometime graphics pictures are displayed by themselves and sometime combined with the actors and live scenes.

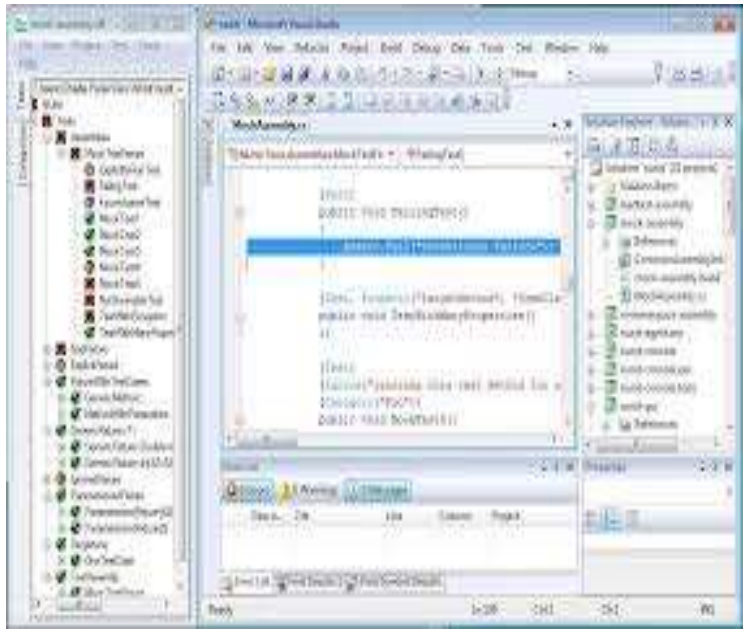
Application Areas of Computer Graphics Medical Field

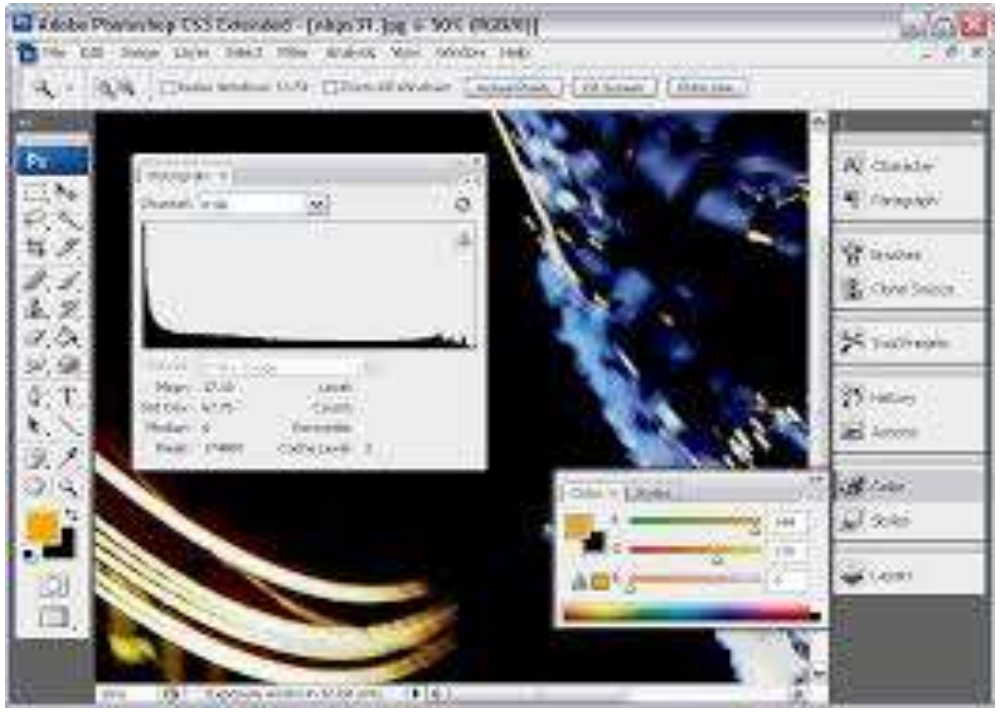
- Computer graphics can also be used to represent the various internal parts and process of the human body.



Application Areas of Computer Graphics Graphical User Interface(GUI)

- It is the interface of the software that communicates with the user with help of some input devices.
  - It contains number of windows , menus and icons for fast selection of processing options (**WIMP**)





Application Areas of Computer Graphics

Graphics Applications Examples

Entertainment: Cinema

Pixar: Monster's Inc.

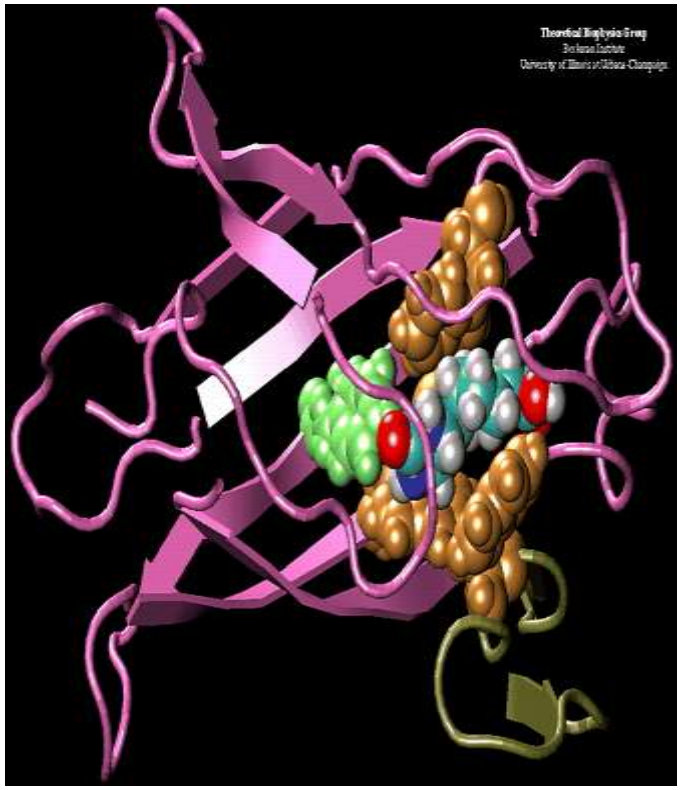
Medical Visualizations

Visible Human Project

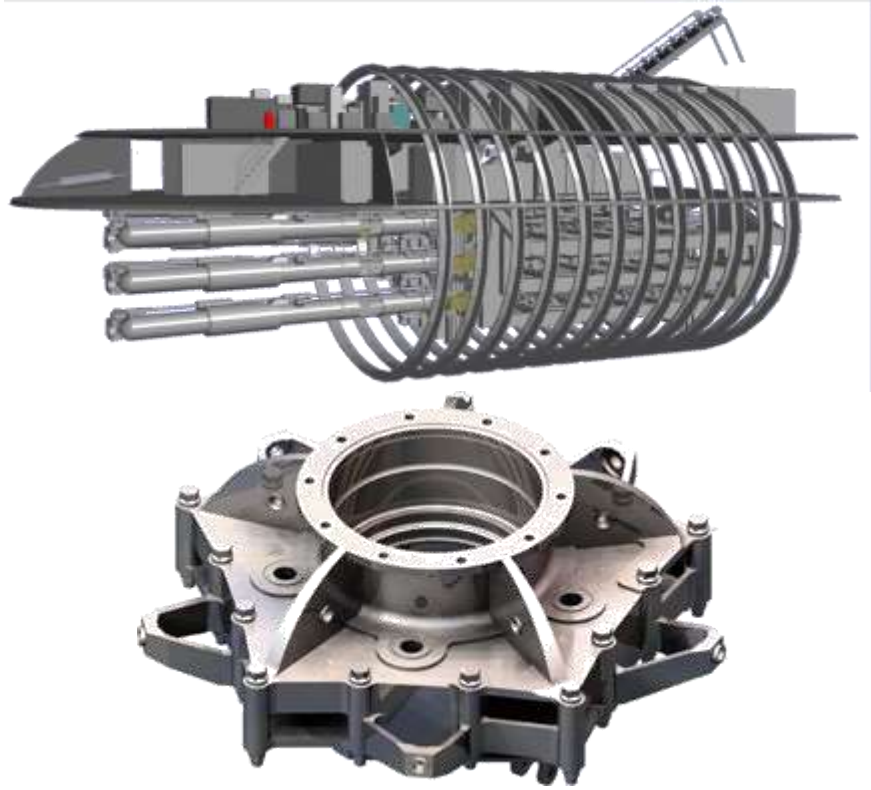
Application Areas of Computer Graphics

Graphics Applications Examples

Scientific Visualization



Computer Aided Design

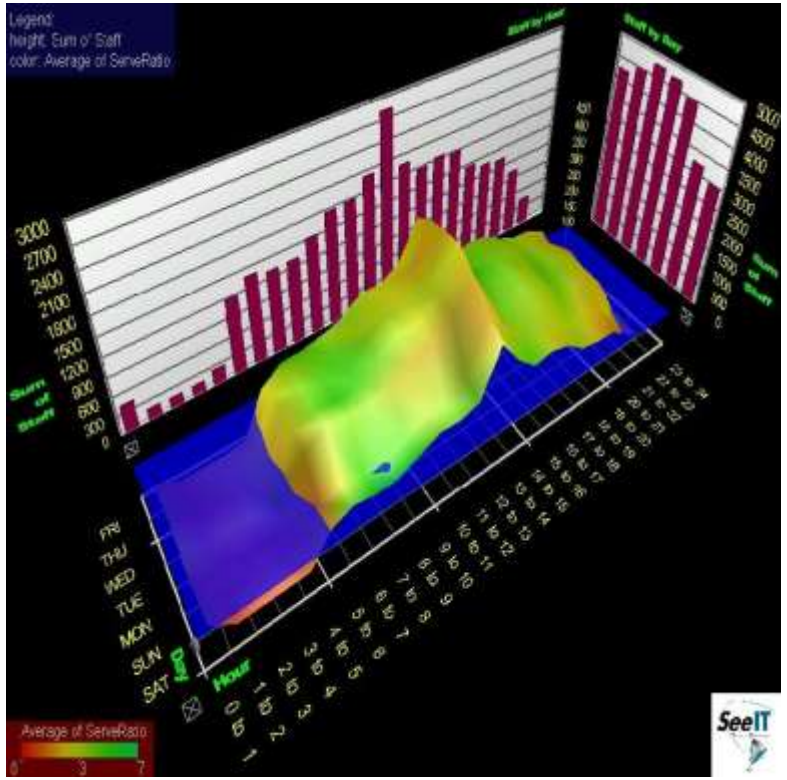


Application Areas of Computer Graphics

Graphics Applications Examples

Entertainment: Games

Information Visualizations



Graphical Display Devices

- Display devices are an output device used for the visual presentation of information for Visual reception.
  1. **Analog display devices (cathode-ray tubes)**
     - Oscilloscope tubes
     - TV CRTs
  2. **Digital display devices**
     - LED (including OLED) displays
     - LCD (liquid crystal) displays
     - VF (vacuum fluorescent) displays
     - Nixie tube displays and PDPs (plasma display panels)
     - Electroluminescent displays (ELDs)
  3. **Others:**
     - Electronic paper, Laser TV etc.

Graphical Display Devices

Cathode Ray Tube (CRT)

- Invented by German physicist Karl Ferdinand Braun in 1897.
  - It is an evacuated glass envelope containing an ***electron gun*** (a source of electrons) and a ***fluorescent screen***, usually with internal or external means to accelerate and deflect the electrons.
  - When electrons strike the fluorescent screen, light is emitted.
  - The electron beam is deflected and modulated in a way which causes it to display an image on the screen.
  - The image may represent ***electrical waveforms*** (***oscilloscope***), ***pictures*** (***television***, ***computer monitor***), echoes of aircraft detected by radar, etc.

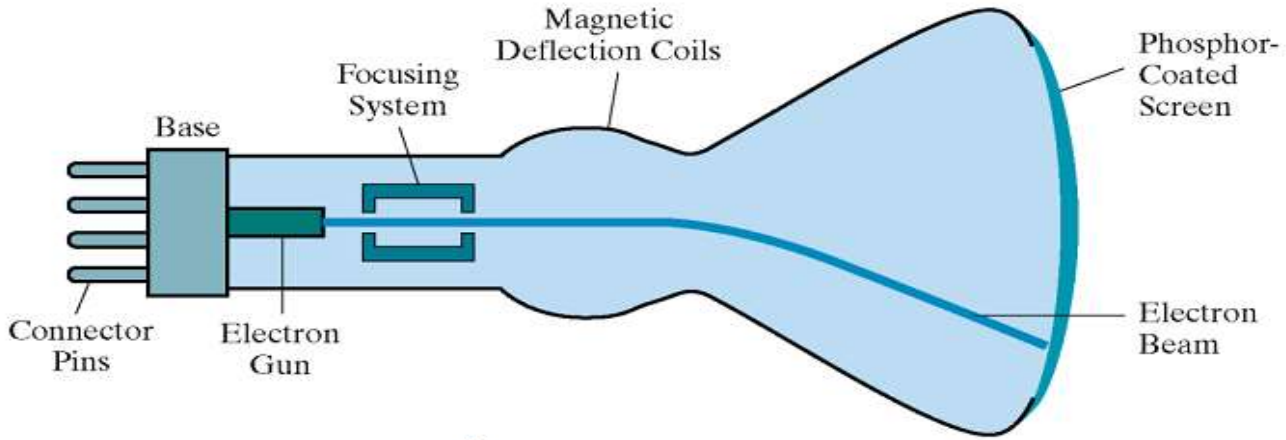
Graphical Display Devices

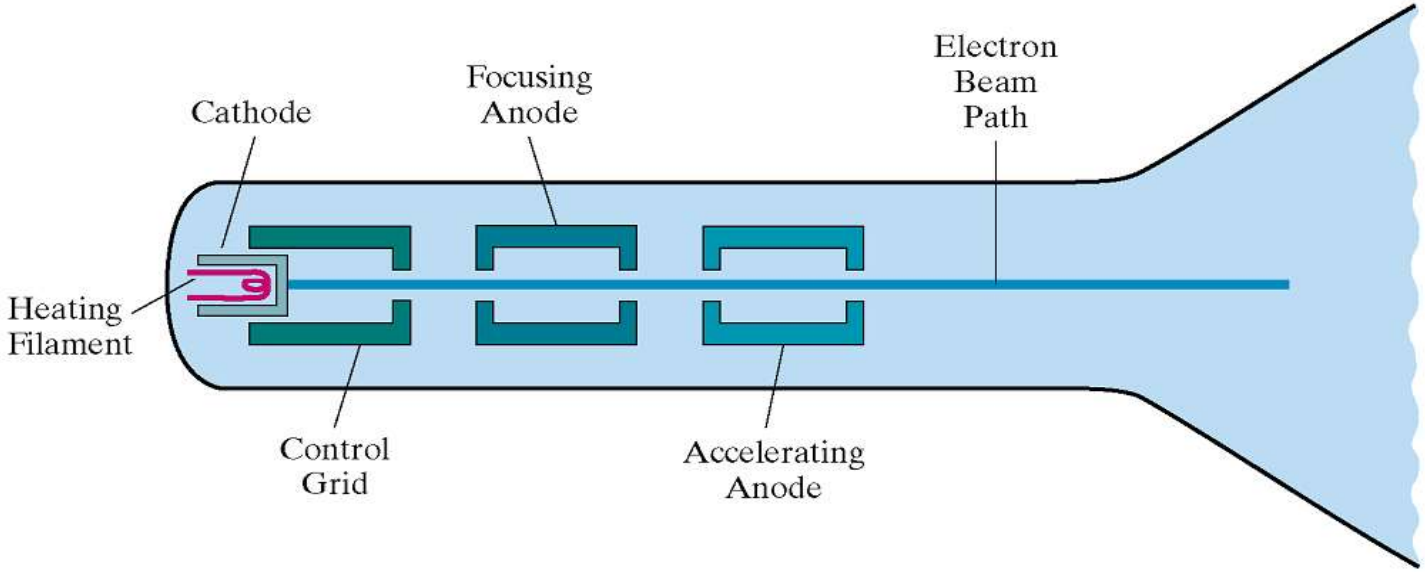
Cathode Ray Tube (CRT)

- A cathode ray tube (CRT) contains four basic parts:
  - ***Electron Gun*** – composed of heated metal cathode and control grid.
  - ***Accelerating Anode***
  - ***Focusing System***
  - ***Deflection system*** (Vertical and horizontal deflection plates)
  - **Phosphor Screen –** evacuated glass envelope/bottle with a phosphorescent screen that glows visibly when struck by the electron beam.

Graphical Display Devices

Cathode Ray Tube (CRT)





Graphical Display Devices

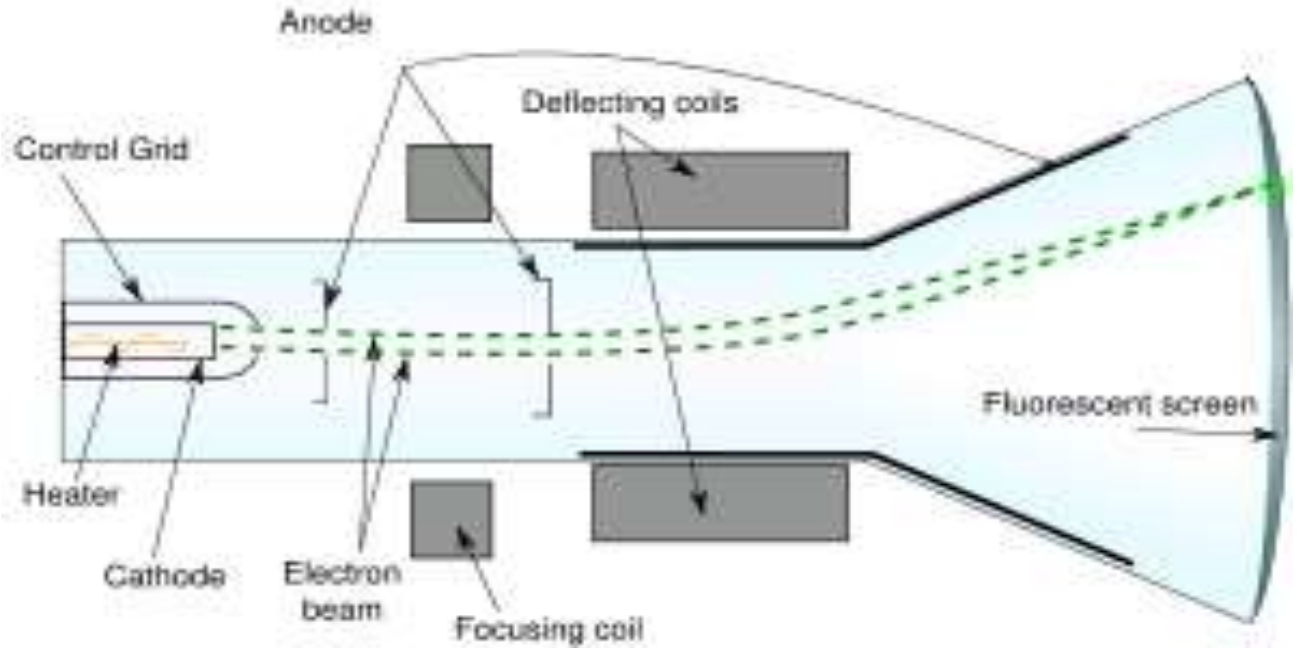
Cathode Ray Tube (CRT)

- **Electron Gun**:
  - An electron gun consists of a series of **electrodes** producing a narrow beam of high-velocity electrons.
  - Electrons are released from the indirectly **heated cathode**.
  - Intensity of the beam is controlled by variation of the negative potential of the cylindrical ***control grid*** surrounding the cathode.
  - This electrode is called the **modulator**.
  - The control grid has a hole in the front to allow passage of the electron beam.
  - The negative voltage applied at cylindrical control grid controls the intensity of electron beam by repealing electrons.
    - High (-ve) voltage stops electron passing from the hole of control grid while small (-ve) voltage decreases electron passage.

Graphical Display Devices

Cathode Ray Tube (CRT)

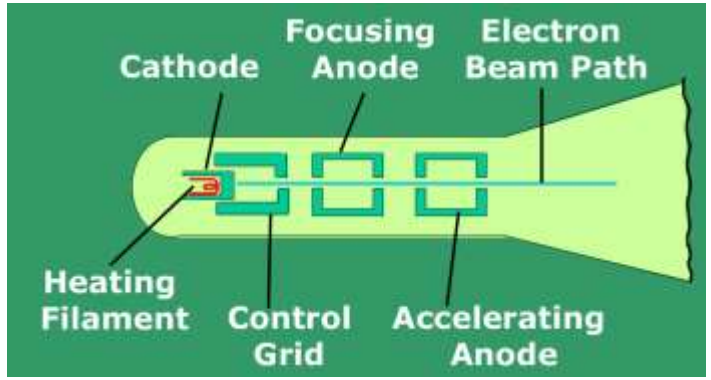
- **Electron Gun**:

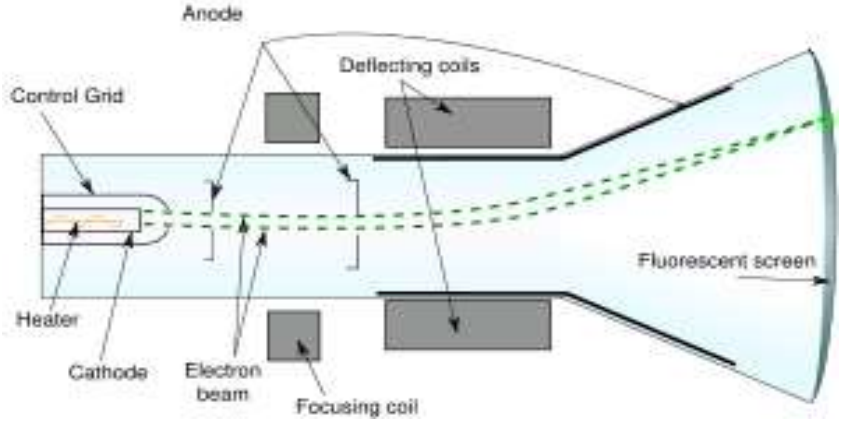


Graphical Display Devices

Cathode Ray Tube (CRT)

- **Focusing System:**
  - In a CRT the focusing system acts like a light lens with a focal length such that the center of focus is the screen.
  - Focusing system concentrates electron beam to a small spot.
    - In ***electrostatic focusing***, electrons pass through positively charged metal cylinder.
    - In ***magnetic focusing***, coils are mounted outside of CRT envelope which produces smallest spot.

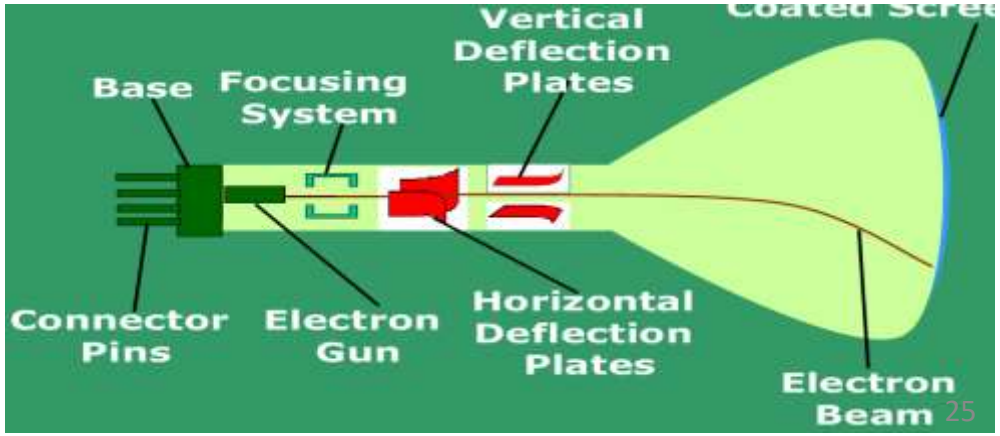




Graphical Display Devices

Cathode Ray Tube (CRT)

- **Deflection System:**
  - Deflection System deflects/directs electron beam horizontally and/or vertically to any point on the screen.
  - The ***horizontal and vertical deflectors*** allow the electron beam to be focused on any spot on the screen.
  - Can be controlled by electric (***deflection plates***) or magnetic fields (***deflection coils***).
    - Magnetic → two pairs of coils
    - Electrostatic → two pairs of deflection plates



Graphical Display Devices

Cathode Ray Tube (CRT)

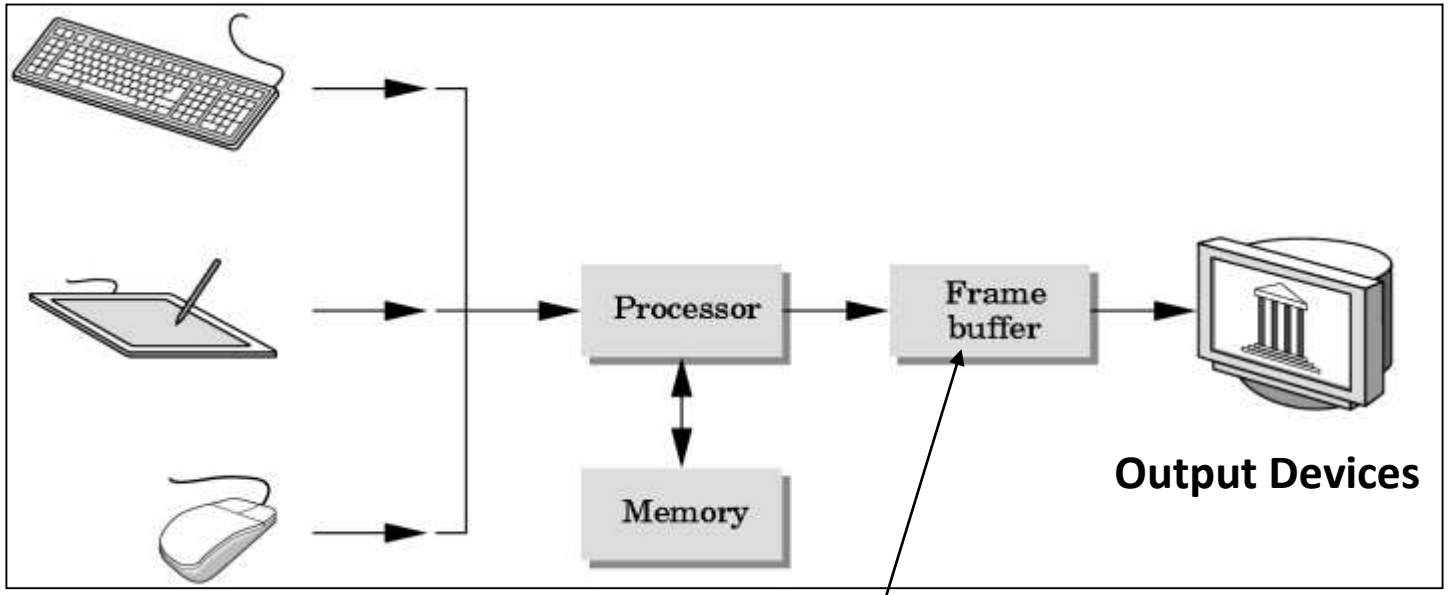
- **Phosphor Screen:**
  - The screen which is the front surface of the picture tube is coated with a special organic compound called a ***phosphor***.
  - Electrons are accelerated towards this phosphor screen with high positive voltage applied at accelerating anode.
  - When the beam hits a phosphor dot, it glows with a brightness proportional to the strength/intensity of the beam and how long it is hit.
  - For color systems there are groups of three different phosphors,
    - One to produce ***red shades***, one for ***green shades***, and one for ***blue shades***.
  - Electrons hit the screen phosphor molecules and cause a ground state to singlet excited state transition.
  - Most of the phosphors relax back to the ground state by emitting a photon of light which is called ***fluorescence***.
    - This happens very rapidly so that all of the molecules which fluoresce do so in under a millisecond.

Graphical Display Devices

CRT Display Technology

- Vector Display
- Raster Display

Basic Graphic System



Output Devices

Input Devices

Image formed in Frame Buffer

Graphical Display Devices

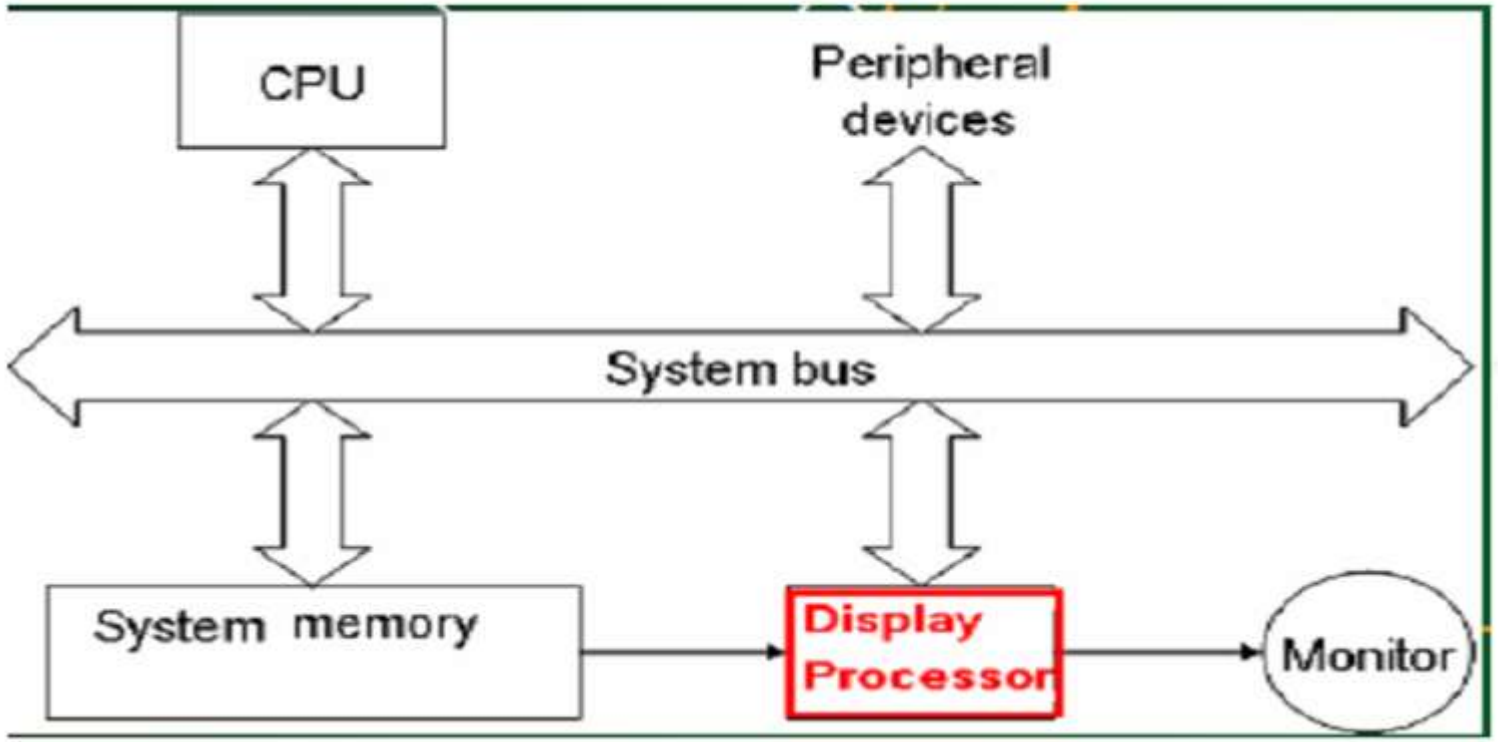
Vector Display Technology

- Often termed **random scan displays**, ***stroke-writing***, or ***calligraphic displays***.
  - Electron beam is directed to the part of screen where picture is to be drawn and directly draws the picture in any specified order.
    - A **pen plotter** is an example of such a system.
  - Picture definition is stored as ***set of line-drawing commands*** in memory, are known as ***refresh display file*** (***refresh buffer/vector file***).
    - The display program (display file) has commands for point, line, and character plotting.
  - The electron gun of a CRT illuminates points and straight lines in any order.
    - To display a Picture, the system cycle through the set of commands in the display file, drawing each component line in turn.

Graphical Display Devices

Vector Display Technology

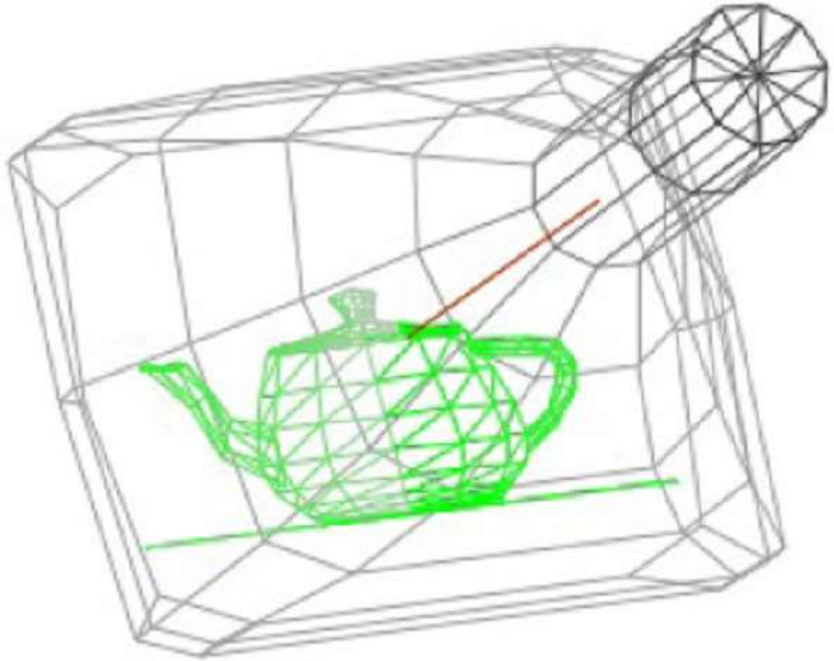
- **Architecture** – Random System

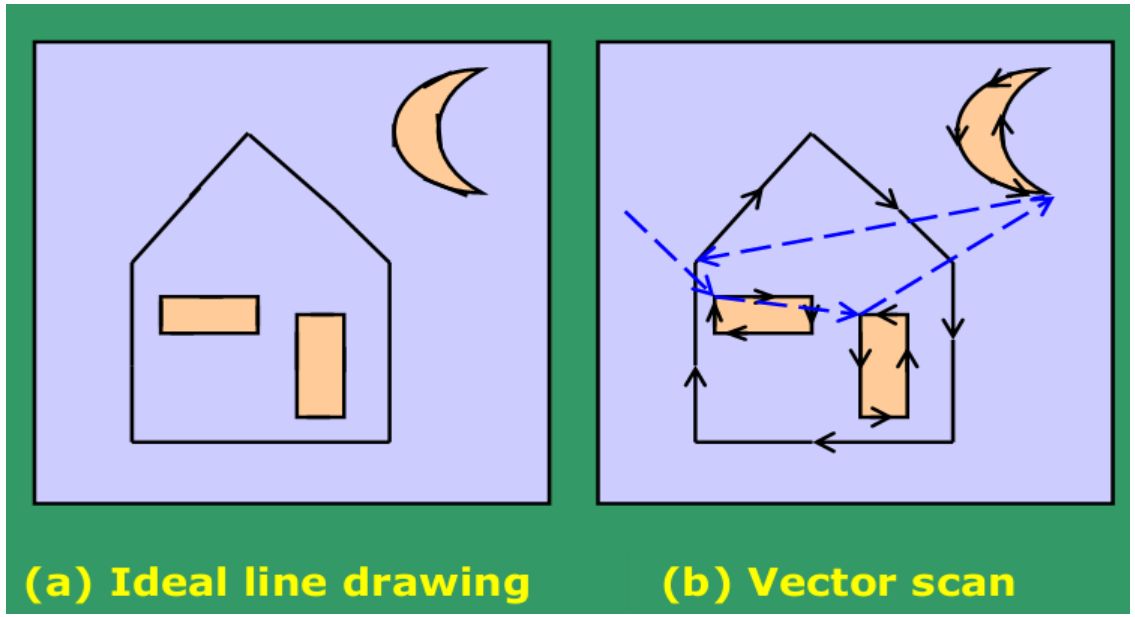


Graphical Display Devices

Vector Display Technology

- Random Scan Displays are designed to draw all the component lines of a picture 30 to 60 times each second.





Graphical Display Devices

Vector Display Technology

- **Advantages**:
  - Very high resolution, limited only by monitor.
  - Easy animation, just draw at different positions.
  - Requires little memory (just enough to hold the display program).
- **Disadvantages**:
  - Requires intelligent electron beam, i.e., processor controlled.
  - Limited screen density before have flicker, can't draw a complex image.
  - Limited color capability (very expensive).
  - Improved in the 1960’s by the **Direct View Storage Tube** (**DVST**) from Tektronix.

Graphical Display Devices

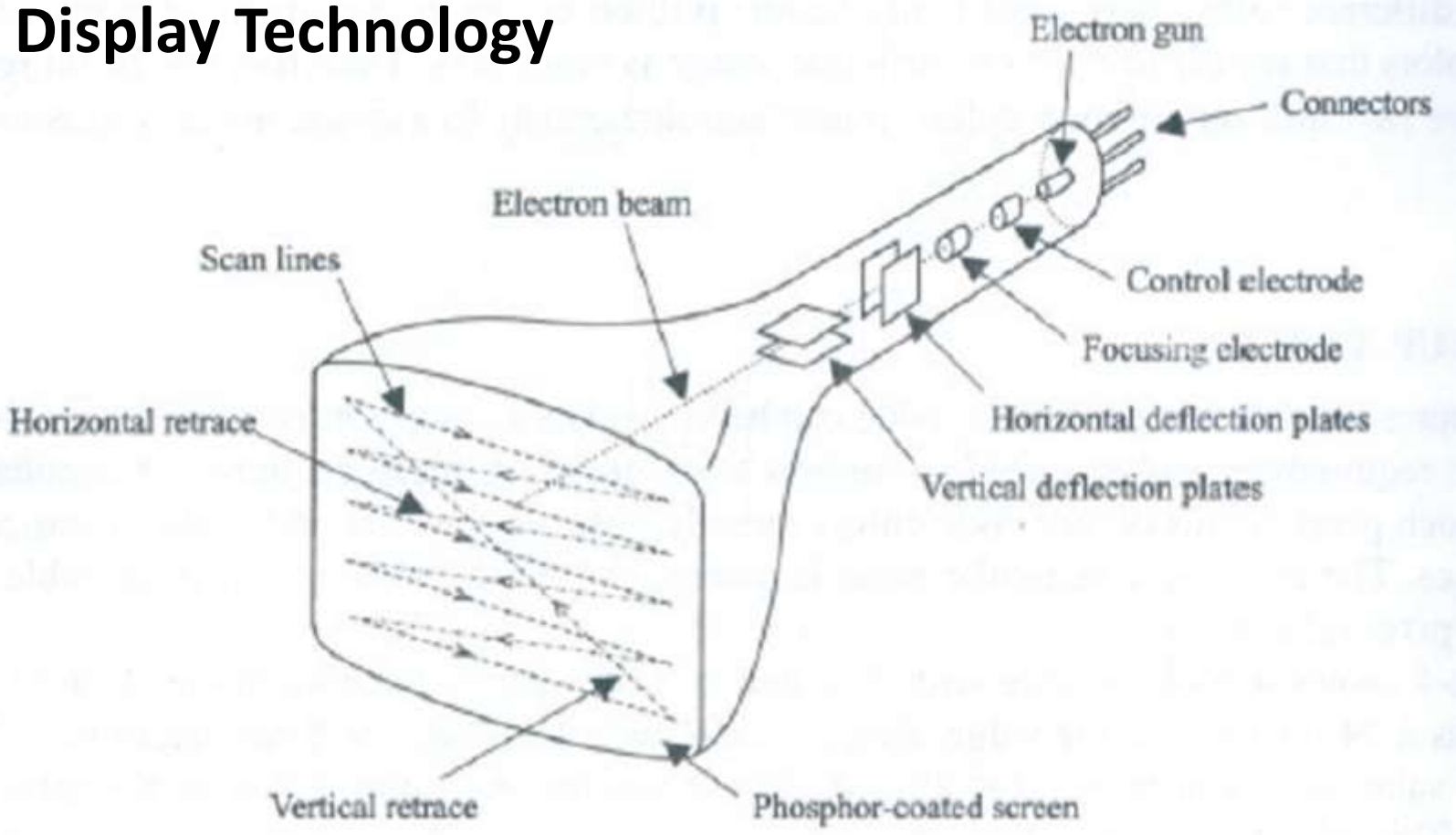
Raster Scan Display Technology

- Raster is a series of adjacent parallel 'lines' which together form an image on a display screen.
  - In early analogue television sets each such line is scanned continuously, not broken up into distinct units.
  - In computer or digital displays these lines are composed of independently colored pixels (picture elements).
- Raster Scan is the representation of images as a collection of ***pixels/dots*** (a rectangular array of points or dots).
  - **Pixel**: one dot or picture element of the raster.
  - **Scan line**: a row of pixels.

- Unlike DVST and random-scan which were line-drawing devices, ***refresh CRT*** is a point-plotting device.
  - Example: Home television sets and printers.

Graphical Display Devices

Raster Scan Display Technology



Graphical Display Devices

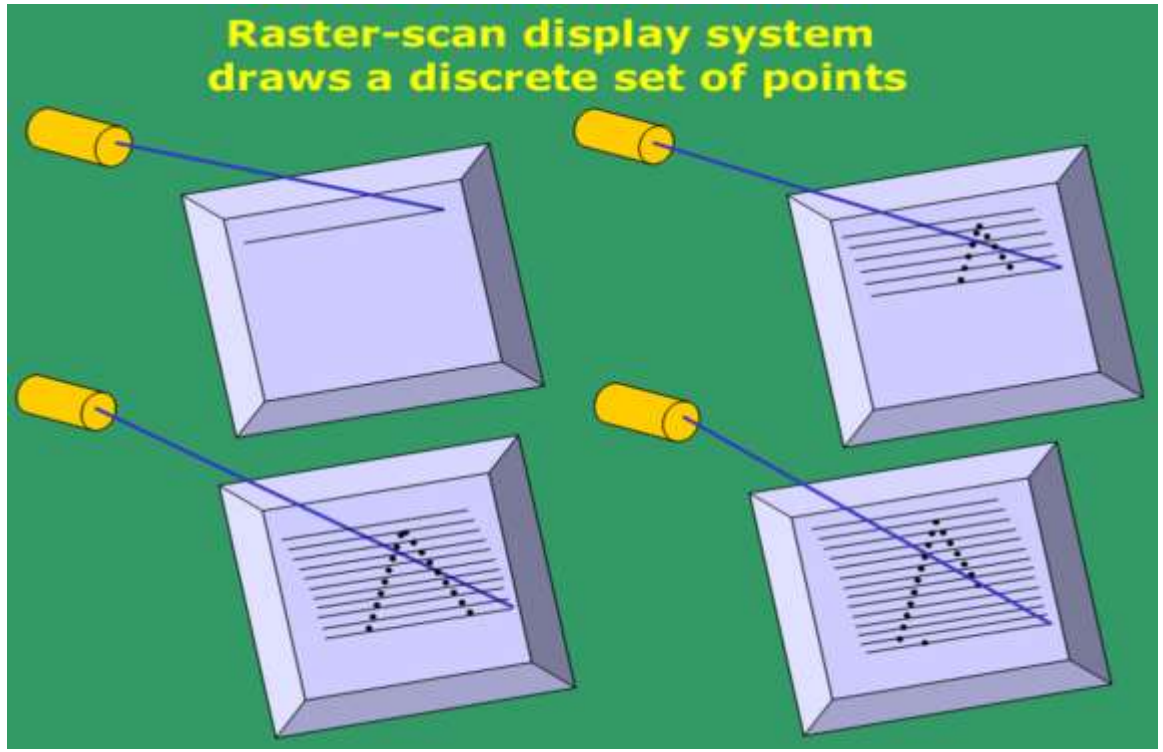
Raster Scan Display Technology – How it works:

- Raster is stored as ***matrix of pixels*** representing entire screen area (i.e. entire screen is a matrix of pixels) and the brightness of each pixel can be controlled by the intensity of the electron beam.
  - **Refresh buffer** can be visualized as a set of horizontal raster lines or a row of individual pixels.
  - Entire image is scanned out sequentially by the video controller (one raster line at a time), i.e. the raster lines are scanned from top to bottom and then back to the top.
  - Line cannot be drawn directly from one point to another, and each point is an addressable point in screen and memory.
    - This causes the effect of '***aliasing***', '***jaggies***' or '***staircase***' effect.
  - At least one memory bit for each pixel (called ***bit-plane***) is required.
  - Refresh/Frame buffer is also called ***Bit-plane***.

Graphical Display Devices

Raster Scan Display Technology – How it works:

- **Refreshing rate** for raster scan display is usually 60 to 80 frames per second.
  - i.e. 1/80 or 1/60 seconds is taken for electron beam to scan from top left corner to bottom right corner.



Graphical Display Devices

Raster Scan Display Technology – How it works:

- **Refresh rate**: 24 is a minimum to avoid flicker, corresponding to 24Hz.
  - Current raster-scan displays have a refresh rate of at least 60 frames (60 Hz) per second, up to 120 (120 Hz).
- **Refresh procedure**:
  - **Horizontal retrace** – beam returns to left of screen.
  - **Vertical retrace** – bean returns to top left corner of screen.
- **Interlaced refresh** – display first even-numbered lines, then odd-numbered lines permits to see the image in half the time
  - Useful for slow refresh rates (30 Hz shows as 60 Hz).
- **Depth of the buffer area** is the number of bits per pixel (***bit planes***), up to 24.

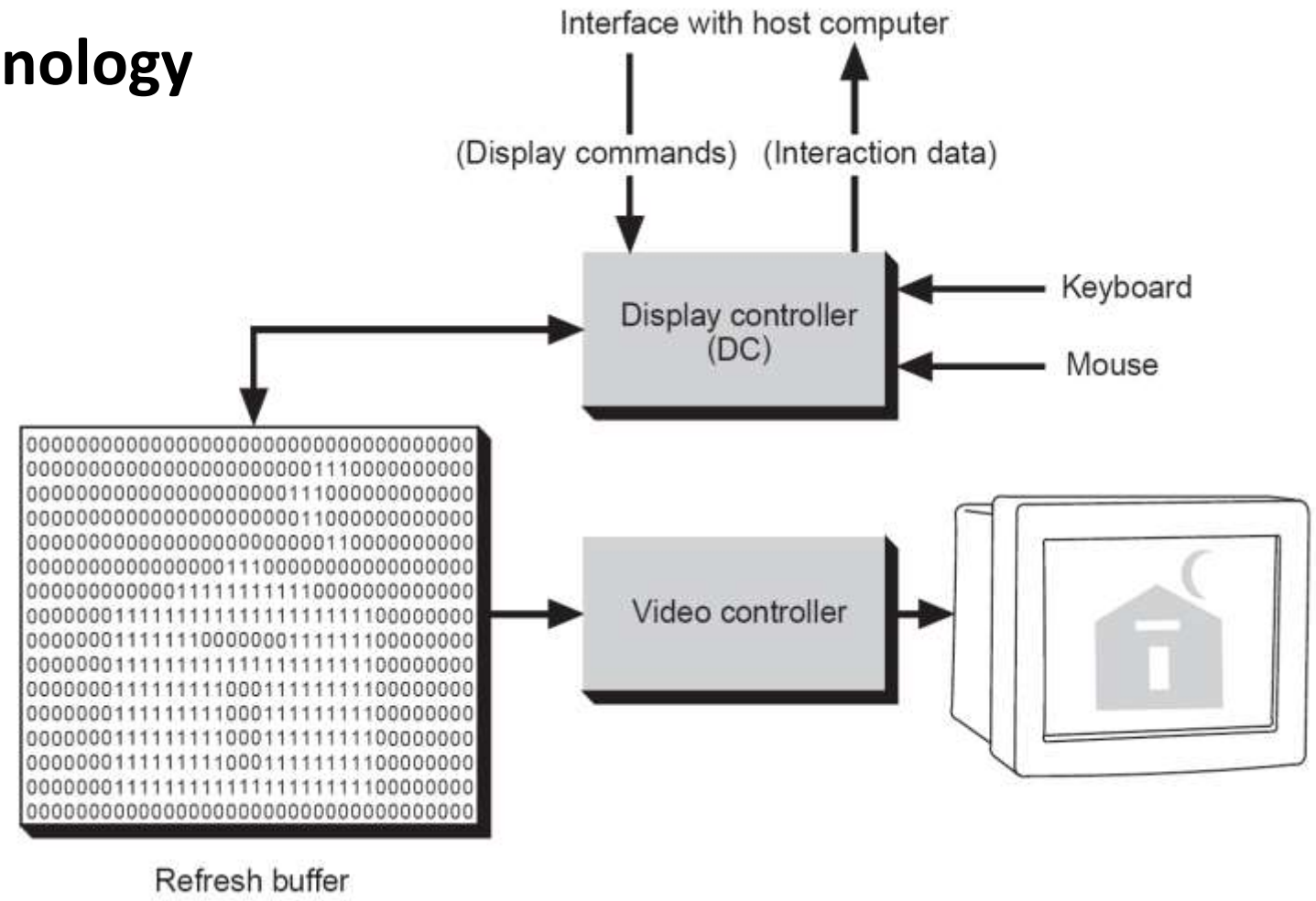
Graphical Display Devices

Raster Scan Display Technology

- In Raster Scan Display
  - A ***display controller*** stores the screen content and other graphics information in a graphics ***memory***.
  - The content of the graphics memory is read by the ***video controller*** at the display's frame rate, converted from digital to analog and transferred to display ***pixel rows*** and ***columns***.

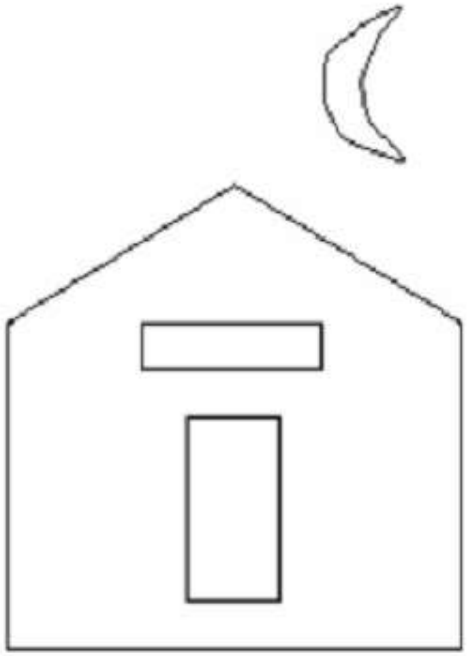
Graphical Display Devices

Raster Scan Display Technology

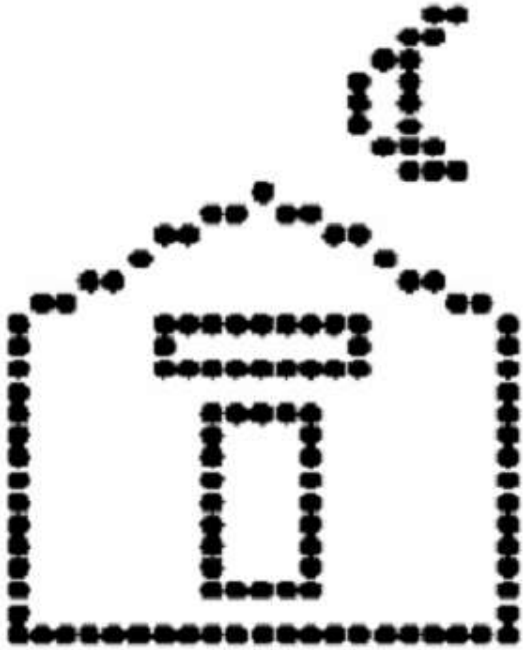


Graphical Display Devices

Raster Scan Display Technology – How it works:



Ideal Drawing

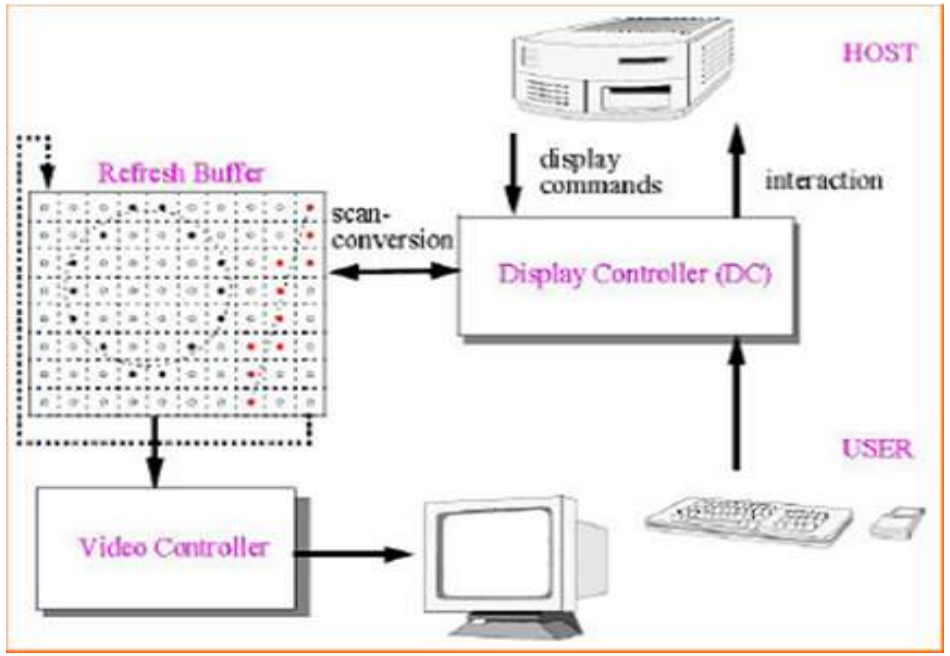


Outline primitives

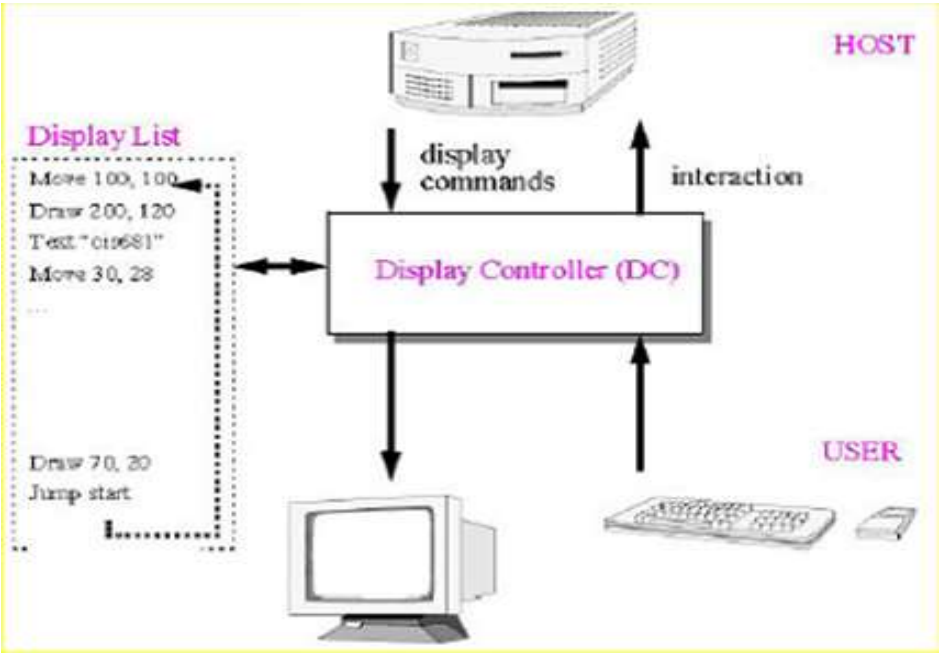
Graphical Display Devices

Raster Scan vs. Random Scan

Raster Scan System



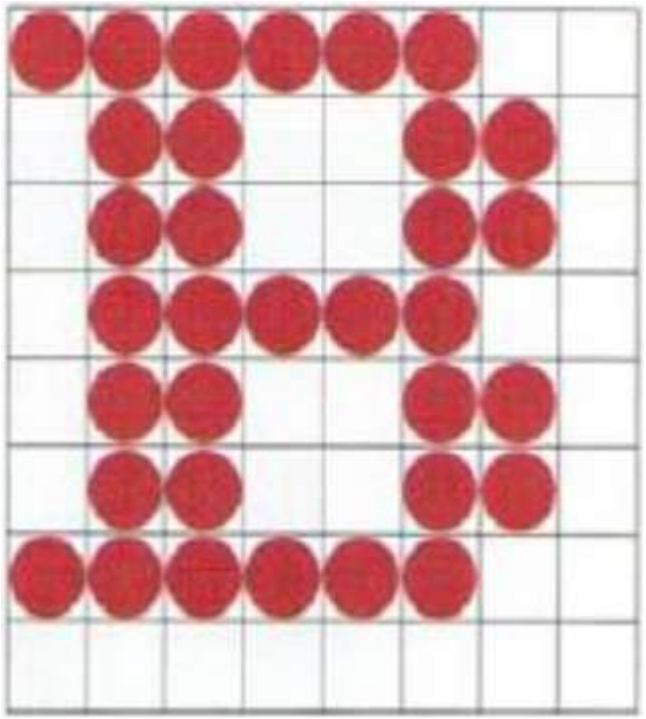
Random Scan System



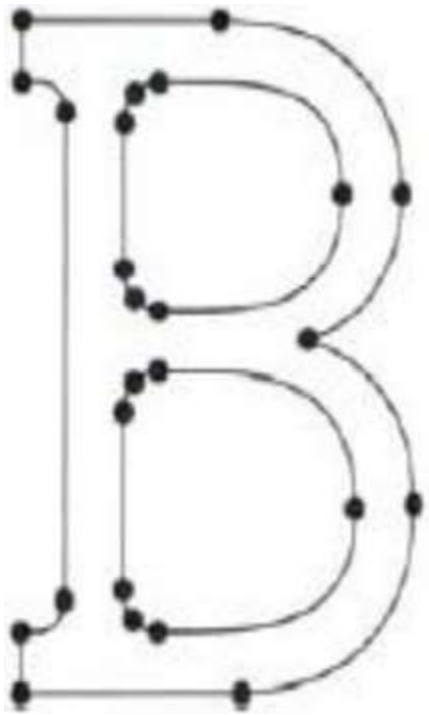
Graphical Display Devices

Raster Scan vs. Random Scan

Raster Scan System



Random Scan System



Graphical Input Devices

- Graphics system make use of different types of input devices for data input. Some of the devices include:
  - Locator Devices
  - Keyboard
  - Scanner
    - Images
    - Laser
  - Cameras (research)

Graphical Input Devices

Locator Devices:

- When queried, locator devices return a position and/or orientation.
  - **Mouse (2D and 3D)**
    - Pointing device to position cursor.
    - Wheel or rollers are used to record the amount and direction of movement.
    - Optical mouse uses optical sensors to detect mouse motion.
    - One, two or three buttons are included.
  - Trackball
  - Joystick (2D and 3D)

Graphical Input Devices

Locator Devices:

- When queried, locator devices return a position and/or orientation.
  - Tablet
  - Virtual Reality Trackers
    - Data Gloves
    - Digitizers

Graphical Input Devices

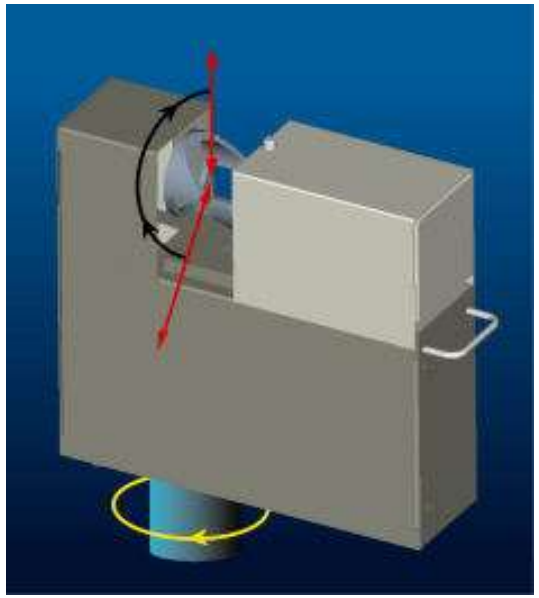
Keyboards:

- **Text input**: ASCII keys are used to input text string.
  - Provides with features to facilitate entry of screen coordinates, menu selections or graphics functions.
  - Function keys allow user to enter frequently used operations in a single stroke and cursor.
  - Control keys are used for cursor position or picture selection.
    - List boxes, GUI
    - CAD/CAM
    - Modeling
- **Hard coded:**
  - Vertex locations are inserted into code.
- Some keyboards consist of trackball or joystick.

Graphical Input Devices

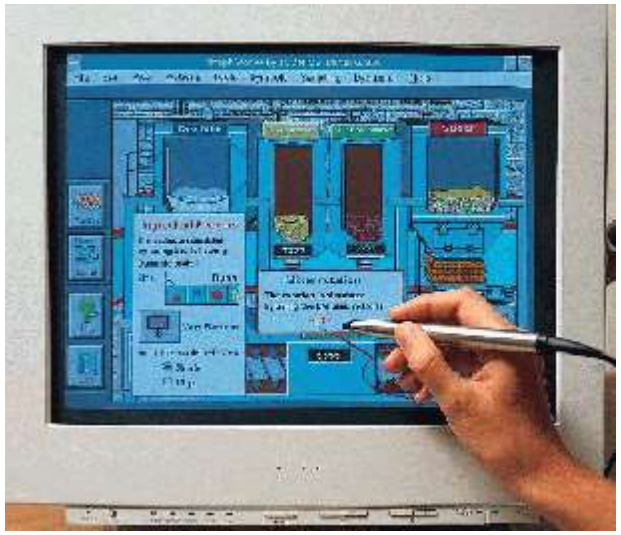
Scanners:

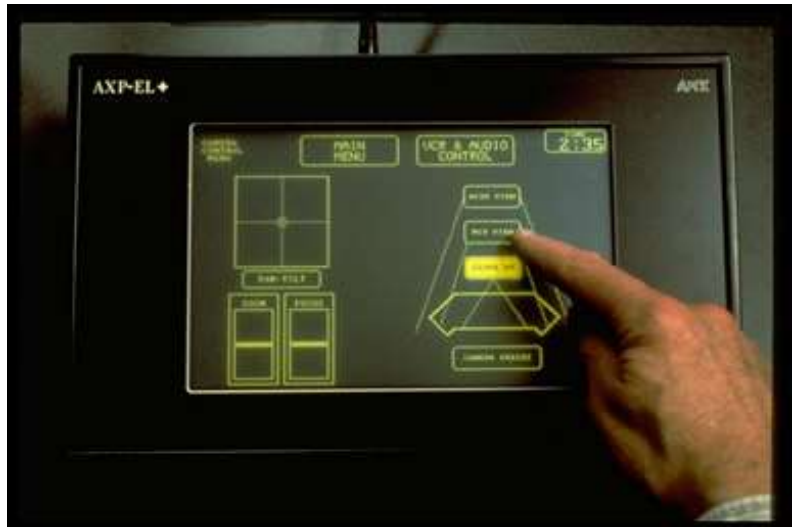
- Image Scanners - Flatbed, etc.
  - The type of data returned is ***Bitmap***.
- Laser Scanners – Deltasphere
  - Emits a laser and does time of flight.
  - Returns 3D point
- Camera based – research
  - Examine camera image(s) and try to figure out vertices from them.



Graphical Input Devices

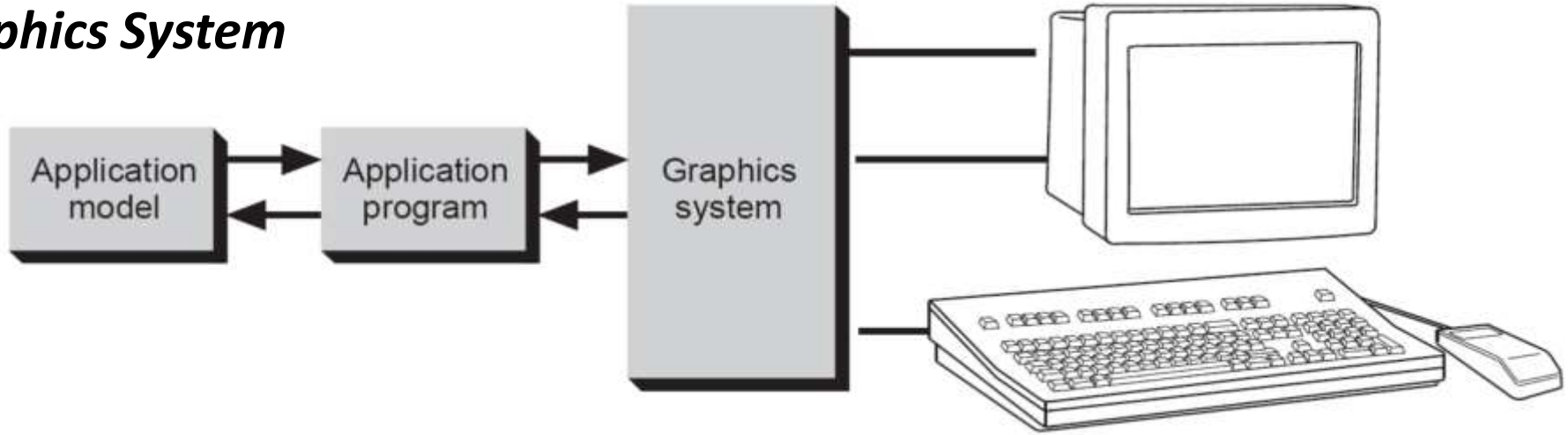
- Many others:
  - Light Pens
  - Voice Systems
  - Touch Panels
  - Camera/Vision Based





Computer Graphics – Systems Architecture

- A computer graphics system consists of the subsequent basic components:
  1. ***Application Model***
  2. ***Application Program***
  3. ***Graphics System***



Computer Graphics – Systems Architecture

1. **Application model**: contains data and objects that are to be displayed on a monitor, such as a scene description or an image file of a vector graphic.
2. **Application Program**: creation of graphical content based on an application model.
   - Transfer of graphical content to the graphics system is by means of graphical output commands.
     - Output commands specify, what is to be shown on the display and how it should be rendered.
   - Storage of graphical content.
   - Example: MS Windows®: classes and methods of the GDI (Graphics Device Interface).

Computer Graphics – Systems Architecture

3. **Graphics system** – the graphics card including ***graphics chip*** and ***memory***:
   - Processing of scene content for display rendering based on application model commands.
   - Transfer of user input commands to the application model.
   - The designer of an application software has to determine
     - which objects and data classes can be generated and
     - which user interactions have to be taken into account.

Computer Graphics – Systems Architecture

Graphics Processing Unit(GPU)

- In the recent years GPUs have evolved into programmable and powerful graphics engines.
- High level parallelism and programmability enable real time computation of complex real world scenes and physical Phenomena.
- Requirements of GPUs are:
  - Huge amounts of data must be processed.
  - Data have to be organized to be processed concurrently.
  - The latency of individual operations is less relevant than the overall throughput.
- GPUs have become GPGPUs (General Purpose GPU).

Computer Graphics – Systems Architecture

Graphics Processing Unit(GPU)

- Example: The GeForce 8800 GTX GPU consists of
    - 8 streaming multiprocessors
    - with 16 stream processors each.
  - The stream processors are also called “unified shaders” and can compute all necessary operations for vertices and pixels.
  - The same operations are performed on multiple data concurrently.
    - Such a computer architecture is called SIMD (Single Instruction, Multiple Data).

Software Standards – GKS, PHIGS, OpenGL

- The primary goal of standardized graphics software is ***portability***.
  - Without standards, programs designed for one hardware system often cannot be transferred to another system without extensive rewriting of the programs.
- The first graphics software standard adopted by ISO and by ANSI is the **General Kernel System (GKS)** in 1984.
  - It was designed as a 2D graphics package, later extended to 3D graphics package.
- The second software standard to be developed and approved by the ISO was **P**rogrammer's **H**ierarchical **I**nteractive **G**raphics **S**ystem (**PHIGS**), which is an extension of GKS.
  - **PHIGS** provided increased capabilities for
    - hierarchical object modeling,
    - color specifications,
    - surface rendering, and
    - picture manipulations
- Subsequently, an extension of PHIGS, called **PHIGS+**, was developed to provide 3D surface-rendering capabilities not available in PHIGS.

Software Standards – GKS, PHIGS, OpenGL

GL and **OpenGL**

- Along with the development of GKS and PHIGS, a set of routines called **GL** (**Graphics Library**), which very soon became a widely used package in the graphics community, and later became a de-facto graphics standard.
- The **GL** routines were designed for *fast*, *real-time rendering*, and soon this package was being extended to other hardware systems.
- As a result, **OpenGL** was developed as a hardware-independent version of **GL** in the early 1990s.
  - This graphics package is now maintained and updated by the OpenGL Architecture Review Board, which is a consortium of representatives from many graphics companies and organizations.
  - The **OpenGL** library is specifically designed for efficient processing of 3D applications, but it can also handle 2D scene descriptions as a special case of three dimensions where all the *z* coordinate values are 0.

End of CH1!

- **Thank You!!!**